

The diagram illustrates the **Data Ecosystem of Robotic Arms**, organized into three main horizontal sections: **Data Acquisition**, **Data Refinement**, and **Data Pipeline**.

Data Acquisition (Left): This section shows the sources of data. **Manipulation Data ($\mathcal{D}$)** includes **Observations** (From Sensors, State transitions $S_t \rightarrow S_{t+1}$ During Execution) and **Contact events** (From End Effectors, Motion sequences From Robotic Arms). **Embodied Datasets** are categorized into **Skill-level Datasets** ($\mathcal{D}_1$), **Task-level Datasets** ($\mathcal{D}_2$), and **Foundation-level Datasets** ($\mathcal{D}_3$). **Real-World Data Collection** involves **VR/AR** and **Handheld** devices, and **UMI** (Upper Limb Interface) and **Teleportation** setups.

Data Refinement (Right): This section shows the process of refining data. It includes **Visual and Viewpoint** refinement, **Physics-Grounded** refinement, **Correction and Recovery** (Base Policy, Motion & Force Corrections), and **Curation and Verification**.

Data Pipeline (Bottom): This section shows the flow of data through various processing stages. It starts with **GR00T-Teleop** (Control Signals, Robot State, Isaac Lab), followed by **GR00T-Mimic** (Motion Annotator, Isaac Lab, Trajectory Generation, Accelerated Physics Engine, Trajectory Evaluator), and finally **GROOT-Gen** (Isaac Sim, 3D, Cosmos).

The diagram also includes a **Benchmark Data Acquisition** section with a **Dataset** visualization and **Automatic Data Generation** using **Ai2THOR**, **MuJoCo**, **CoppeliaSim**, **ManiSkill**, and **NVIDIA**.

The diagram illustrates the Data Pipeline, which is divided into three main stages: GR00T-Teleop, GR00T-Mimic, and GROOT-Gen.

- GR00T-Teleop:** A user (represented by a person icon) provides a **Teleop Demonstration** (represented by a hand icon) to **Isaac Lab** (represented by a green ring icon). **Isaac Lab** outputs **Control Signals** (represented by a stack of coins icon) and **Robot State** (represented by a stack of coins icon).
- GR00T-Mimic:** A **Motion Annotator** (represented by a green cube icon) provides data to **Isaac Lab** (represented by a green ring icon). **Isaac Lab** then feeds into a **Trajectory Generation** module (represented by a gear icon), which is part of an **Accelerated Physics Engine** (represented by a cube icon with a wrench). The output of the **Trajectory Generation** module is fed into a **Trajectory Evaluator** (represented by a gear icon).
- GROOT-Gen:** The output of the **Trajectory Evaluator** is fed into **Isaac Sim** (represented by a green ring icon), which outputs **3D** data (represented by a green cube icon). This 3D data is then processed by **Cosmos** (represented by a cube icon with a brain).

